HAO ZHOU

hao.zhou@utdallas.edu HAI Lab ECSS 4.410, UT Dallas

ACADEMIC APPOINTMENT

Assistant Professor 2026.08 -
Department of Computer Science, University of Texas at Dallas

RESEARCH AREAS

AI-Powered Mobile and Wearable Systems · Human-Centered AI · AI for Health

EDUCATION

The Pennsylvania State University
Doctor of Philosophy in Computer Science and Engineering 2021.08 - 2026.05

The University of Mississippi
Master of Science in Computer Science 2019.08 - 2021.05
Bachelor of Science in Computer Science 2016.08 - 2019.05

HONORS & AWARDS

2025	Penn State Alumni Association Dissertation Award
2025	Samsung Research America President Award
2025	ACM MobiSys Rising Stars
2025	UIST Special Recognition for Belonging and Inclusion Award
2024	ACM MobiCom Student Travel Grant
2023	Entrepreneurial Lead in NSF I-Corps Program
2023	ACM/IEEE IoTDI Best Paper Award for Edge IoT AI
2023	Microsoft Research Asia Stars of Tomorrow
2023	Penn State International Student Travel Grant
2022	Penn State Outstanding Teaching Assistant Award

GRANTS

2026 - 2027 **PI;** Samsung Research America ($150,000).

PUBLICATIONS

ICML 26 *Physiology-Aware Masked Cross-Modal Reconstruction for Biosignal Representation Learning.*
Hao Zhou, Simon A. Lee, Cyrus Tanade, Keum San Chun, Juhyeon Lee, Migyeong Gwak, Megha Thukral, Justin Sung, Eugene Hwang, Mehrab Bin Morshed, Li Zhu, Viswam Nathan, Md Mahbubur Rahman, Subramaniam Venkatraman, Sharanya Arcot Desai

MobiCom 26 *Exploring the Feasibility of Full-Body Muscle Activation Sensing with Insole Pressure Sensors.*
Hao Zhou, Mahanth Gowda

ICLR 26 *HiMAE: Hierarchical Masked Autoencoders Discover Resolution Specific Structure in Wearable Time Series.*
Simon Lee, Cyrus Tanade, **Hao Zhou**, Juhyeon Lee, Megha Thukral, Minji Han, Rachel Choi, Md Sazzad Hissain Khan, Baiying Lu, Migyeong Gwak, Mehrab Bin Morshed, Viswam Nathan, Md Mahbubur Rahman, Li Zhu, Subramaniam Venkatraman, Sharanya Arcot Desai

Nature Comm. under review *ECG-informed pretraining enables precise cardiac assessment from wearable photoplethysmography.*
Hao Zhou, et al.

Nature Comm. under review *A compact on-device foundation model enables real-time physiological screening on wearable devices.*
Simon A. Lee, **Hao Zhou**, et al.

NeurIPS 26 TS4H *Towards On-device Foundation Models for Wearable Signals.*
Simon Lee, Cyrus Tanade, **Hao Zhou**, Juhyeon Lee, Megha Thukral, Minji Han, Baiying Lu, Sharanya Desai

ICASSP 26 *A Personalized Real-time Proactive Voice Memory Assistant.*
Hao Zhou, Md Mahbubur Rahman, Simon A Lee, Baiying Lu, Juhyeon Lee, Cyrus Tanade, Megha Thukral, Md Sazzad Hissain Khan, Samsad Ul Islam, Subramaniam Venkatraman, Sharanya Arcot Desai

MobiSys 25 Rising Star *Rethinking Inexpensive Wearables in the Era of AI: From Motion Analytics to Mobile Health.*
Hao Zhou

ICASSP 25 *Know Your Heart Better: Multimodal Cardiac Output Monitoring Using Earbuds.*
Hao Zhou, Md Mahbubur Rahman, Mehrab Bin Morshed, Yunzhi Li, Md Saiful Islam, Larry Zhang, Jungmok Bae, Christina Rosa, Wendy Berry Mendes, Jilong Kuang

UbiComp 25 *WindDancer: Understanding Acoustic Sensing under Ambient Airflow.*
Kuang Yuan, Dong Li, **Hao Zhou**, Zhehao Li, Lili Qiu, Swarun Kumar, Jie Xiong

UIST 25 *SignGlass: First-Person View Comprehensive and Generalizable ASL Translation Using Wearable Glasses.*
Yongxiang Cai, Taiting Lu, Zhenghao Li, **Hao Zhou**, Kenneth DeHaan, Xuhai Xu, Mahanth Gowda, Yincheng Jin

CHI 25 *BallistoBud: Heart Rate Variability Monitoring Using Earbud Accelerometry for Stress Assessment.*
Md Saiful Islam, Md Mahbubur Rahman, Mehrab Bin Morshed, David J. Lin, Yunzhi Li, **Hao Zhou**, Wendy Berry Mendes, and Jilong Kuang

MobiCom 24 *Rethinking Orientation Estimation with Smartphone-Equipped Ultra-Wideband Chips.*
Hao Zhou, Kuang Yuan, Mahanth Gowda, Lili Qiu, Jie Xiong

IoTDI 24 *ASLRing: American Sign Language Recognition with Meta-Learning on Wearables.*
Hao Zhou, Taiting Lu, Kenneth DeHaan, Mahanth Gowda

UbiComp 24 *SmartDampener: An Open Source Platform for Sport Analytics in Tennis.*
Runze Liu, Taiting Lu, Shengming Yuan, **Hao Zhou**, Mahanth Gowda

MobiCom 23 *SignQuery: A Natural User Interface and Search Engine for Sign Languages with Wearable Sensors.*
Hao Zhou, Taiting Lu, Kristina Mckinnie, Joseph Palagano, Kenneth DeHaan, Mahanth Gowda

IoTDI 23 *One Ring to Rule Them All: An Open Source SmartRing Platform for Finger Motion Analytics and Healthcare Applications.*
Hao Zhou, Taiting Lu, Yilin Liu, Shijia Zhang, Runze Liu, Mahanth Gowda

TIOT 23 *I Am an Earphone and I Can Hear My User's Face: Facial Landmark Tracking Using Smart Earphones.*
Shijia Zhang, Taiting Lu, **Hao Zhou**, Yilin Liu, Runze Liu, Mahanth Gowda

NeurIPS 23 Workshop *Backdoor Threats from Compromised Foundation Models to Federated Learning.*
Xi Li, Songhe Wang, Chen Wu, **Hao Zhou**, Jiaqi Wang

UbiComp 22 *Learning on the Rings: Self-Supervised 3D Finger Motion Tracking Using Wearable Sensors.*
Hao Zhou, Taiting Lu, Yilin Liu, Shijia Zhang, Mahanth Gowda

INDUSTRY EXPERIENCE

2025.07 - 2026.07 Digital Health Team, Samsung Research America

2024.05 - 2024.08 Digital Health Team, Samsung Research America

2023.05 - 2023.08 Microsoft Research Asia - Shanghai

PATENTS

2026 Pending — *Cross-Modality Encoding and Reconstruction of Related Physiological Signals.* **Hao Zhou**, Md Mahbubur Rahman, Cyrus Tanade, Subbu Venkatraman, Sharanya Desai

2026 Pending — *A Personalized Real-time Proactive Voice Memory Assistant.* **Hao Zhou**, Md Mahbubur Rahman, Subbu Venkatraman, Sharanya Desai

2026 Pending — *Efficient On-Device Learning Frameworks for Wearable Sensor Signals.* Simon A. Lee, **Hao Zhou**, Minji Han, Rachel Choi, Juhyeon Lee, Sharanya Desai

2025 — *Cardiovascular Biomarker Estimation Using PPG And BCG from a Wearable Device.* Mehrab Bin Morshed, Md Mahbubur Rahman, Li Zhu, **Hao Zhou**, Larry Zhang

2024 — *Platform for finger motion analytics and healthcare applications.* Taiting Lu, **Hao Zhou**, Mahanth Gowda, Yilin Liu, Shijia Zhang, Runze Liu

MEDIA COVERAGE

2026 — xMAE: Samsung News / Machine Brief / RichlyAI

2025 — Penn State Alumni Association Dissertation Award: Penn State News

2023 — OmniRing: Hackster / DeepTech

INVITED TALKS

2026.10 — Auburn CS

2026.03 — UW-Madison CS

2026.02 — UT Dallas CS

2026.02 — UMBC CSEE

2026.02 — UAB CS

2025.11 — Penn State EE

2023.05 — Microsoft Research Asia

TEACHING

Instructor @ UT Dallas

- CS 4390: Computer Networks — SP 2027
- CS 4485: Computer Science Project — FA 2026

Teaching Assistant @ Penn State

- CMPEN/EE 362: Communication Networks — SP 2025

- CSE/EE 559: Wireless and Mobile Sensing in the age of IoT — FA 2024
- CMPEN 462: Wireless Communication Systems and Security — SP 2022, SP 2024

SERVICE

Program Committee

- ACM MobiCom S3 Workshop — 2024
- ACM MobiCom Artifact Evaluation — 2024
- ACM MobiSys Artifact Evaluation — 2024

Invited Reviewer

- ACM UbiComp/IMWUT — 2022 - 2027
- IEEE Transactions of Mobile Computing — 2024 - 2027
- npj Nature Digital Medicine — 2026 - 2027
- ACM CHI — 2026 - 2027
- ICLR — 2024 - 2027
- ICML — 2026 - 2027
- NeurIPS — 2026 - 2027
- TMLR — 2026 - 2027
- AAAI — 2027
- ACM Transactions on Sensor Networks — 2024
- Journal of Intelligent Manufacturing — 2022